2.2 Proposal for a Further Empirical Design

This section proposes a new experimental design to mitigate these negative externalities based on the theoretical analysis. Drawing on Lowe's (2021) distinction between "collaborative contact" and general interaction, this design introduces a goal-oriented collaborative task as an intervention. The aim is to examine if the shared team identity established through collaborative contact can remedy the backlash triggered by affirmative action and thereby restore the coordination advantage of minority leaders.

2.2.1 Research Gap: Mitigating the Backlash Effect

Although existing literature has separately explored the effects of affirmative action and intergroup contact, there has been no systematic examination of their combined effects. Bhalotra et al. (2023, pp. 10-11) designed an experiment in which "affirmative action" and "intergroup contact" were set as independent intervention conditions. While this design separately revealed that AA triggered a backlash from the majority group, thereby reducing efficiency (Bhalotra et al., 2023, p. 19), and intergroup contact could generally enhance the coordination level of both groups (Bhalotra et al., 2023, pp. 20-21), it still remains a key policy design gap: Can contact serve as a remedial mechanism specifically to counteract the negative externalities brought about by AA? This constitutes the core research gap that this paper aims to fill.

To address this gap, a more refined contact mechanism must be introduced. Lowe (2021, p. 2) distinguished between "collaborative contact" and " competitive contact " in an experiment in India and pointed out that only when group members work towards "common goals" can contact effectively reduce prejudice and promote cross-group interactions (Lowe, 2021, pp. 7-8). Theoretically, this collaborative experience based on common goals can establish a "teammate" bond that transcends identities; when this bond is strong enough, it may psychologically reduce the stigmatizing signals associated with introducing a quota system, thereby suppressing backlash.

Based on the above theoretical reasoning, this paper proposes the following two core hypotheses:

Hypothesis 0 (The Backlash Effect): In the absence of prior collaborative contact, the introduction of AA (quotas) will lead the majority group (Hindus) to reduce cooperation with minority group (Muslim) leaders, thereby reducing coordination efficiency. This hypothesis aims to replicate the efficiency paradox observed by Bhalotra et al. (2023, p. 20).

Hypothesis 1 (The Mitigation Effect): After introducing "collaborative contact" as a prior intervention, the backlash effect triggered by AA will be significantly weakened. Specifically, groups that have experienced collaborative tasks will show significantly higher coordination efficiency when facing minority "quota leaders" compared to those that have not experienced collaboration.

2.2.2 Experimental Setup: Integrating collaborative Contact

To test the above hypothesis, this section designs a controlled laboratory experiment with three treatment groups. This design retained Bhalotra et al.'s (2023, p. 9) "weakest-link" coordination game as the core measurement tool, while introducing different intervention stages before the game began.

The core innovation of this design lies in the strict distinction of “contact types". Although Bhalotra et al. (2023, p. 15) used a puzzle task as "intergroup contact" in their experiment, the main purpose was to promote communication and familiarity among participants of different religions. In contrast, Lowe (2021, p. 11) emphasized that the core of "collaborative contact" is not just interaction, but "cooperative striving for the goal". To separate these two mechanisms and differentiate "familiarity" and the deeper "teammate identity," this study set up the following three experimental groups:

1. Baseline Group (AA Only): This group aims to replicate the efficiency paradox observed by Bhalotra et al. Participants were informed that the leader was selected based on a quota, but there was no form of interaction before the coordination game began. This serves as a comparison benchmark to establish the existence of the backlash effect.
2. Treatment 1 (AA + Social Contact): This group aims to control the "familiarity" effect. Before the quota rules were announced and the game began, group members engaged in 10

minutes of unstructured social interaction (such as self- introductions and ice-breaking topic discussions), but did not involve any common task goals or rewards. This design simulates ordinary social contact and aims to test whether merely eliminating strangeness is sufficient to overcome the backlash.

3. Treatment 2 (AA + Collaborative Contact): This group is the core intervention group of this study, aiming to test the collaborative mechanism proposed by Lowe (2021, p. 4). Before the leader was selected, groups (each containing more than 2 persons) had to complete a high-difficulty logic puzzle task (similar to Bhalotra's jigsaw puzzle, but with strong incentives). The key is that this task was set up as an "all-or-nothing" reward mechanism: only if the group succeeded together could all members receive additional bonuses. This design forcibly builds a temporary community through a common goal with members from different religions, thereby pre-establishing a "teammate" identity at the psychological level in order to counteract the stigmatizing signal brought by the introduction of AA.

One of the core challenges in the design of this study lies in how to distinguish the essential differences between "ordinary social contact" and "cooperative contact". Scacco and Warren (2018, p. 674) provided important clues from their research in the conflict zone of Kaduna, Nigeria: The research found that even in a highly isolated and conflict-ridden environment, social contact based on skills training, although it failed to significantly change the participants' prejudice towards their group, significantly reduced discriminatory behavior. This means that relying on the ordinary social interaction (Social Contact) in Treatment 1 may not be sufficient to reverse the deep psychological stereotypes. Therefore, a more effective intervention mechanism must be introduced. Lowe (2021, p. 4) clearly distinguished between "adversarial contact" and "collaborative contact" in the cricket league experiment in India. He found that only when members of different castes are in the same team and strive for the common goal of victory (i.e., collaborative contact), cross-group friendships and trade efficiency will increase; conversely, adversarial contact will instead intensify hostility. Lowe’s (2021, p. 21) data indicate that this contact based on a common goal can increase cross-caste friendships by 35%. Based on this，Treatment 2 of this study particularly emphasizes the "all-or-nothing" incentive mechanism. This is not only to create contact opportunities, but also to use the "common goal" mechanism described by Lowe before the quota announcement to psychologically reconstruct the identity of " Teammate ", thereby building a firewall to resist the backlash effect brought by the quota in the future.

2.3 Procedural Details and Expected Outcomes

This section outlines the specific implementation parameters, benefit function settings, and measurement analysis framework of the proposed experiment. To ensure the comparability of the results, the experimental parameters strictly follow the standards set by Bhalotra et al. (2023, p. 9).

The experiment plans to recruit 240 college students as participants and randomly assign them to 60 groups. According to the setting of Bhalotra et al. (2023, pp. 8-9), each group consists of 4 members, including 2 majority group members and 2 minority group members. These 60 groups will be further randomly and evenly divided into three experimental conditions (20 groups in each condition): (1) Baseline (AA Only), (2) Treatment 1 (Social Contact), (3) Treatment 2 (Collaborative Contact).

The experimental process is divided into three consecutive stages. The first stage introduce and proceed intervention, where each group performs different pre-game tasks: The Baseline group does not involve interaction; The Groups in Treatment 1 conducts the duration of unstructured social interaction; The groups in Treatment 2 need to jointly solve a difficult logical puzzle within 15 minutes and implement the "all-or-nothing" (all-or-nothing) bonus distribution rule to build functional dependence. Then comes stage two with quota announcement, where the experimenter announces that the leadership position is reserved for the minority group based on the quota of AA, in order to activate the backlash psychology. Stage three carries out the main task under the leadership of the quota leader for 6 rounds of "the weakest link" coordination game.

In the game, each member i chooses the effort level:

$$x_i \in \{0, 10, \ldots, 100\}$$

The benefit function is consistent with that of Bhalotra et al. (2021, p. 9):

$$\pi_i = 500 - 25x_i + [\ \backslash\min(x_1, \ldots, x_4) \times 40\ ]$$

This formula indicates that the individual benefit is determined by the minimum group effort level (the weakest link). Since any effort higher than the group's minimum value will lead to a loss in individual benefit, the realization of an efficient equilibrium highly depends on the reactions of members towards the identity of the leader.

To quantify the impact of collaborative contact on mitigating the backlash effect, we will conduct an analysis using the following regression model (adapted from Bhalotra et al., 2023, p. 15):

$$\mathrm{MinGroupEffort}_g = \alpha + \beta_1 \cdot T_{\{Social\}} + \beta_2 \cdot T_{\{Collaborative\}} + X_g\gamma + \epsilon_g$$

where $\mathrm{MinGroupEffort}_g$represents the average minimum effort level of the group g. Based on Lowe (2021, p. 11)'s finding that a common goal can reduce bias, we expect $\beta_1 \approx 0$ (compared to Baseline group), indicating that merely relying on ordinary social contact cannot eliminate the stigma brought about by quotas; at the same time, we expect $\beta_2 > 0$ and to be significant, suggesting that collaborative contact successfully establishes trust across identities, thereby counteracting the negative effects of affirmative action.

The summarized experimental setup and expected outcomes sees following Table:

Condition	Pre-game interaction	Common goal & joint incentive	Expected outcome
Baseline (AA only)	None	No	Strong majority backlash; low minimum group effort, minority leaders lose baseline advantage
Treatment 1 (AA + social contact)	Unstructured social interaction (e.g. introductions, chatting)	No	Backlash largely unchanged; minimum effort similar to baseline.
Treatment 2 (AA + collaborative contact)	Joint puzzle task in mixed groups	Yes (all-or-nothing group bonus)	Backlash mitigated; higher minimum effort and restored coordination under minority quota leaders.